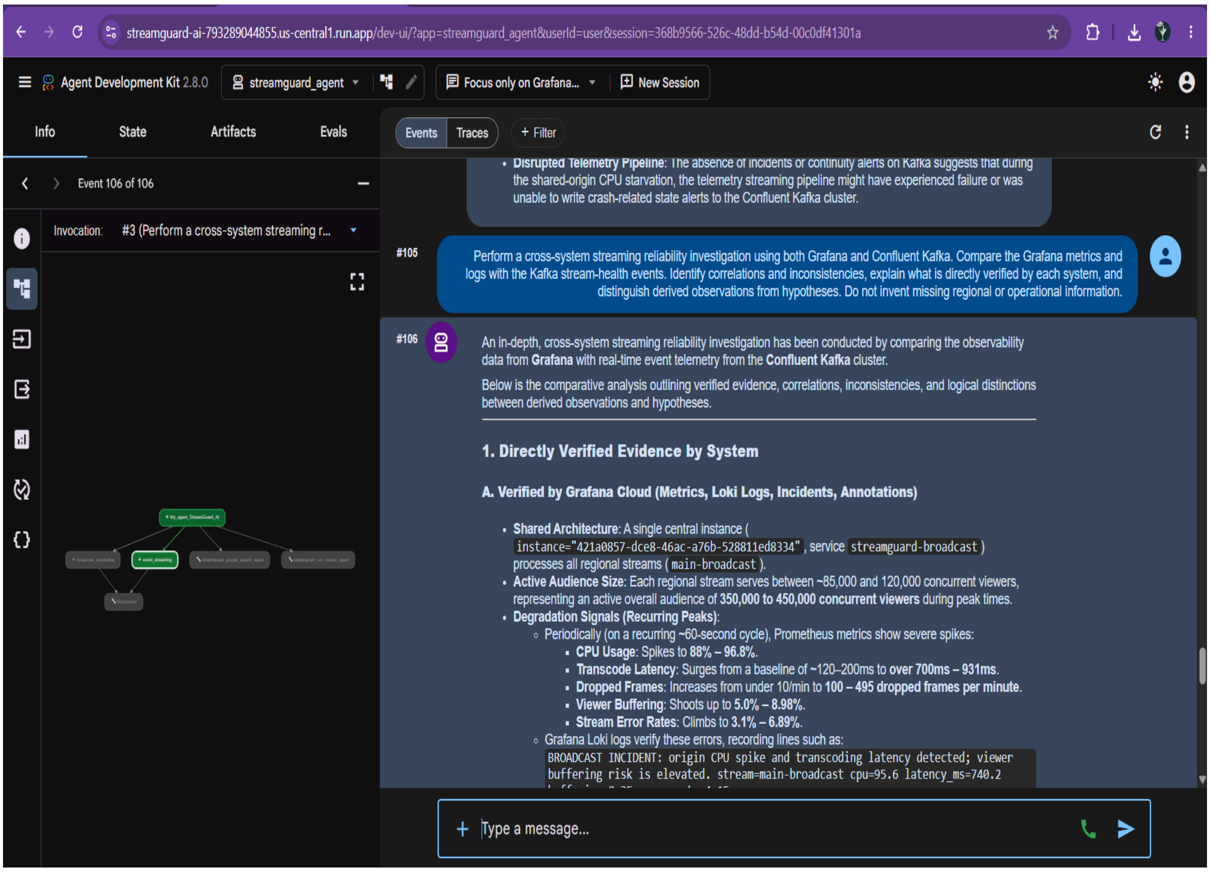

streamguard-ai-793289044855.us-central1.run.app/dev-ui/?app=streamguard_agent&userId=user&session=368b9566-526c-48dd-b54d-00c0df41301a
Agent Development Kit 2.8.0
streamguard_agent
Focus only on Grafana...
New Session
Info
State
Artifacts
Evals
Event 106 of 106
Invocation:
#3 (Perform a cross-system streaming r...
Events
Traces
+ Filter
• Disrupted Telemetry Pipeline: The absence of incidents or continuity alerts on Kafka suggests that during the shared-origin CPU starvation, the telemetry streaming pipeline might have experienced failure or was unable to write crash-related state alerts to the Confluent Kafka cluster.
#105
Perform a cross-system streaming reliability investigation using both Grafana and Confluent Kafka. Compare the Grafana metrics and logs with the Kafka stream-health events. Identify correlations and inconsistencies, explain what is directly verified by each system, and distinguish derived observations from hypotheses. Do not invent missing regional or operational information.
#106
An in-depth, cross-system streaming reliability investigation has been conducted by comparing the observability data from Grafana with real-time event telemetry from the Confluent Kafka cluster.
Below is the comparative analysis outlining verified evidence, correlations, inconsistencies, and logical distinctions between derived observations and hypotheses.
1. Directly Verified Evidence by System
A. Verified by Grafana Cloud (Metrics, Loki Logs, Incidents, Annotations)
• Shared Architecture: A single central instance (instance="421a0857-dce8-46ac-a76b-528811ed8334" , service streamguard-broadcast) processes all regional streams (main-broadcast).
• Active Audience Size: Each regional stream serves between ~85,000 and 120,000 concurrent viewers, representing an active overall audience of 350,000 to 450,000 concurrent viewers during peak times.
• Degradation Signals (Recurring Peaks):
◦ Periodically (on a recurring ~60-second cycle), Prometheus metrics show severe spikes:
▪ CPU Usage: Spikes to 88% – 96.8%.
▪ Transcode Latency: Surges from a baseline of ~120–200ms to over 700ms – 931ms.
▪ Dropped Frames: Increases from under 10/min to 100 – 495 dropped frames per minute.
▪ Viewer Buffering: Shoots up to 5.0% – 8.98%.
▪ Stream Error Rates: Climbs to 3.1% – 6.89%.
◦ Grafana Loki logs verify these errors, recording lines such as:
BROADCAST INCIDENT: origin CPU spike and transcoding latency detected; viewer buffering risk is elevated. stream=main-broadcast cpu=95.6 latency_ms=740.2
Type a message...